ZK-KGVerify: Privacy-Preserving Verification of Knowledge Graph Reasoning using Zero-Knowledge Proofs and Blockchain

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***Abstract*—Knowledge graph (KG) reasoning powered by deep learning has achieved remarkable success in tasks such as link prediction and relation inference. However, deploying these models in high-stakes domains—healthcare, supply chain, finance—raises critical challenges around *verifiability* and *privacy*: How can a third party verify that a prediction originated from a legitimately trained model, without the model owner revealing proprietary weights or sensitive training data? We propose ZK-KGVerify, a novel framework that bridges knowledge graph reasoning with zero-knowledge proofs (ZKPs) and blockchain technology. Our system trains graph neural network (GNN)-based KG embedding models, generates Pedersen commitment-based zero-knowledge proofs attesting that predictions were computed from committed model parameters, and logs verification results on a blockchain for an immutable audit trail. We evaluate three KG embedding models—TransE, RotatE, and CompGCN—on the FB15k-237 benchmark, and analyze the overhead introduced by the ZKP and blockchain layers. Our results demonstrate that ZK-KGVerify achieves 100% verification and tamper detection with $\sim$1 ms proof generation overhead, making privacy-preserving verification of KG reasoning practical for real-world deployment.**

***Index Terms*—Knowledge Graphs, Graph Neural Networks, Zero-Knowledge Proofs, Blockchain, Link Prediction, Privacy-Preserving Machine Learning, Verifiable AI**

I. Introduction

Knowledge graphs (KGs) have emerged as a powerful formalism for representing structured knowledge, encoding entities and their relationships as directed graphs. When combined with deep learning—particularly graph neural networks (GNNs)—KG reasoning enables automated inference of missing links, anomaly detection, and relational reasoning across domains ranging from drug discovery [1] to financial fraud detection [2].

Motivating example. Consider a pharmaceutical company that has built a proprietary knowledge graph linking drugs, proteins, diseases, and side-effects. A GNN model trained on this KG predicts that *Drug X treats Disease Y*—a commercially valuable inference. A hospital wishes to act on this prediction but needs assurance that it came from a properly trained model, not a random guess. The pharmaceutical company, in turn, cannot share model weights (trade secrets) or training data (patient records protected by HIPAA/GDPR). ZK-KGVerify resolves this impasse: the company proves the prediction is genuine via a zero-knowledge proof, and the hospital verifies it without learning any proprietary information. A blockchain provides a tamper-proof audit trail of all verified predictions.

In simple terms, ZK-KGVerify works as follows: (1) a GNN model is trained on a knowledge graph and makes predictions; (2) the model owner creates a cryptographic "fingerprint" (Pedersen commitment) of the prediction's internal data without revealing it; (3) a zero-knowledge proof convinces any third party that the fingerprint is genuine; and (4) the verification result is permanently recorded on a blockchain.

However, deploying KG reasoning models in production environments introduces a fundamental trust trilemma:

(i) **Verifiability**: A downstream consumer of model predictions needs assurance that the prediction was generated by a properly trained model, not fabricated or computed by an inferior model.
(ii) **Privacy**: The model owner may not wish to reveal model weights (proprietary intellectual property) or training data (potentially containing sensitive information subject to regulations such as GDPR).
(iii) **Immutability**: There must be a tamper-proof record that a particular prediction was verified at a specific time, preventing retroactive alteration of audit logs.

Existing approaches address these concerns in isolation. Federated learning [3] preserves data privacy but does not provide verifiable proofs. Model watermarking [4] offers ownership verification but leaks model information. Blockchain-based ML systems [5] provide immutability but lack privacy guarantees.

In this paper, we propose **ZK-KGVerify**, an end-to-end framework that simultaneously addresses all three challenges. Our key contributions are:

- A **novel architecture** integrating GNN-based KG embeddings, Pedersen commitment-based zero-knowledge proofs, and blockchain-anchored verification into a unified pipeline.

- A **ZKP protocol** based on Schnorr-type sigma protocols with Fiat-Shamir heuristic that proves a prediction was computed from committed model parameters without revealing the parameters themselves.
- A **comprehensive evaluation** of three KG embedding models (TransE, RotatE, CompGCN) on FB15k-237, including link prediction accuracy, ZKP overhead analysis, and blockchain gas cost estimation.
- An **open-source implementation**[1] that runs entirely on a single GPU (Colab T4), making the research reproducible.

The remainder of this paper is organized as follows. Section II reviews related work. Section III introduces mathematical preliminaries. Section IV details the ZK-KGVerify framework. Section V presents experimental setup and results. Section VI provides analysis and discussion. Section VII concludes with future directions.

II. Related Work

A. Knowledge Graph Embeddings

Knowledge graph embedding methods learn low-dimensional representations of entities and relations. **TransE** [6] models relations as translations ($\mathbf{h} + \mathbf{r} \approx \mathbf{t}$), while **RotatE** [7] extends this to complex space, modeling relations as rotations that capture symmetry, antisymmetry, and composition patterns. GNN-based approaches such as R-GCN [8] and **CompGCN** [9] leverage message passing over the graph structure, jointly learning entity and relation embeddings through composition operators for richer relational modeling.

B. Zero-Knowledge Proofs in Machine Learning

Zero-knowledge proofs (ZKPs) enable a prover to convince a verifier of a statement's truth without revealing any additional information [10]. Recent work has applied ZKPs to machine learning: zkML [11] and EZKL [12] enable ZK verification of neural network inference by compiling computation graphs into arithmetic circuits. **zkCNN** [13] provides zero-knowledge proofs for convolutional neural network predictions with sumcheck-based protocols. **vCNN** [14] offers verifiable convolutional network predictions using homomorphic commitments. Ghodsi et al. [**?**] proposed proof-of-inference protocols for general DNN models, demonstrating that verifiable inference is feasible at practical latencies.

However, these approaches target feedforward or convolutional networks and do not address graph-structured models or knowledge graph reasoning specifically. ZK-KGVerify fills this gap by applying ZKP verification to GNN-based KG embeddings—a model family with unique challenges including variable-size neighborhoods, relation-typed edges, and message-passing aggregation.

[1]Code will be released upon publication.

C. Blockchain for Trustworthy AI

Blockchain technology provides decentralized, immutable ledgers [15], [16]. Prior work has explored blockchain for ML model marketplaces [5], federated learning coordination [17], and training data provenance [18]. DInEMMo [19] uses blockchain for model management but lacks privacy-preserving verification. Our work uniquely combines blockchain immutability with ZKP privacy for knowledge graph reasoning.

III. Preliminaries

A. Knowledge Graphs and Link Prediction

A knowledge graph $\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{T})$ consists of a set of entities $\mathcal{E}$, relations $\mathcal{R}$, and triples $\mathcal{T} \subseteq \mathcal{E} \times \mathcal{R} \times \mathcal{E}$. Each triple $(h, r, t) \in \mathcal{T}$ represents a fact: entity h (head) is related to entity t (tail) via relation r.

Link prediction is the task of predicting missing triples. Given $(h, r, ?)$, the model ranks all candidate entities $t' \in \mathcal{E}$ by a scoring function $f(h, r, t')$.

B. KG Embedding Models

1) TransE: TransE [6] embeds entities and relations into $\mathbb{R}^d$ and defines the scoring function as the negative L2 distance after translation:

$$f_{\text{TransE}}(h, r, t) = -\|\mathbf{e}_h + \mathbf{e}_r - \mathbf{e}_t\|_2 \tag{1}$$

where $\mathbf{e}_h, \mathbf{e}_r, \mathbf{e}_t \in \mathbb{R}^d$ are the embedding vectors of head, relation, and tail respectively.

2) RotatE: RotatE [7] models relations as rotations in complex space $\mathbb{C}^d$:

$$f_{\text{RotatE}}(h, r, t) = -\|\mathbf{e}_h \circ \mathbf{e}_r - \mathbf{e}_t\|_2 \tag{2}$$

where $\circ$ denotes element-wise product and $\mathbf{e}_r = e^{i\boldsymbol{\theta}_r}$ with learned phase angles $\boldsymbol{\theta}_r \in \mathbb{R}^d$, enabling the model to capture symmetry, antisymmetry, inversion, and composition patterns.

3) CompGCN: CompGCN [9] performs composition-based aggregation:

$$\mathbf{e}_i^{(l+1)} = \sigma\left(\sum_{(j,r)\in\mathcal{N}(i)} \mathbf{W}_{\lambda(r)}^{(l)} \phi(\mathbf{e}_j^{(l)}, \mathbf{e}_r^{(l)})\right) \tag{3}$$

where $\phi(\cdot,\cdot)$ is a composition operator (subtraction, multiplication, or circular correlation), $\lambda(r)$ indicates the direction (original, inverse, or self-loop), and relation embeddings are jointly updated:

$$\mathbf{e}_r^{(l+1)} = \mathbf{W}_{\text{rel}}^{(l)} \mathbf{e}_r^{(l)} \tag{4}$$

C. Pedersen Commitments

A Pedersen commitment scheme [20] allows a party to commit to a value v while keeping it hidden, with the ability to reveal it later. Given generators $g, h \in \mathbb{G}$ of a group of prime order p where the discrete logarithm $\log_g h$ is unknown:

$$\mathsf{Commit}(v, r) = g^v \cdot h^r \mod p \tag{5}$$

where $r \xleftarrow{\$} \mathbb{Z}_p$ is a random blinding factor. Pedersen commitments are:

- **Hiding**: C reveals no information about v (computationally or perfectly, depending on variant).
- **Binding**: The committer cannot find $v' \neq v, r'$ such that $\mathsf{Commit}(v', r') = C$ (computationally binding under discrete log assumption).
- **Homomorphic**: $\mathsf{Commit}(v_1, r_1) \cdot \mathsf{Commit}(v_2, r_2) = \mathsf{Commit}(v_1 + v_2, r_1 + r_2)$.

D. Schnorr Identification and Fiat-Shamir Heuristic

The Schnorr protocol [21] is a three-move sigma protocol for proving knowledge of a discrete logarithm. Combined with the Fiat-Shamir heuristic [22], it becomes non-interactive:

1) **Commitment**: Prover selects random $k \xleftarrow{\$} \mathbb{Z}_p$, computes announcement $A = g^k$.
2) **Challenge**: $e = \mathsf{Hash}(A \| \text{statement})$ (Fiat-Shamir).
3) **Response**: $s = k + e \cdot x \mod p$ where x is the secret.
4) **Verification**: Check $g^s \stackrel{?}{=} A \cdot y^e$ where $y = g^x$.

E. Blockchain and Immutable Audit Trails

A blockchain is a distributed, append-only ledger where blocks are linked via cryptographic hashes [15]. Each block B_i contains a set of transactions $\mathcal{TX}_i$ and includes $\mathsf{Hash}(B_{i-1})$, the hash of the previous block. This hash-chain structure ensures that altering any historical block invalidates all subsequent blocks, providing tamper-evident storage. Smart contracts are self-executing programs deployed on the blockchain that enforce verification logic without a trusted intermediary [16]. In ZK-KGVerify, we use a blockchain to create an immutable record of which predictions were verified and when, enabling retrospective auditing.

IV. ZK-KGVerify Framework

A. System Overview

ZK-KGVerify consists of three integrated layers, illustrated in Figure 1:

1) **KG Reasoning Layer**: Trains GNN-based models on a knowledge graph for link prediction.
2) **ZKP Verification Layer**: Generates and verifies zero-knowledge proofs that predictions originate from committed model parameters.
3) **Blockchain Audit Layer**: Records verification outcomes as immutable on-chain transactions.

B. Threat Model

We consider three parties:

- **Model Owner (Prover)**: Possesses a trained KG embedding model $\mathcal{M}$ with parameters θ. Wants to serve predictions without revealing θ.
- **Consumer (Verifier)**: Receives a prediction $(h, r, \hat{t})$ and wants assurance it came from a legitimately trained model.
- **Adversary**: May attempt to (a) serve predictions from an untrained/random model, (b) tamper with proofs after generation, or (c) alter the audit log.

Security goals: (i) *Soundness*—a dishonest prover cannot produce a valid proof for a prediction not computed from the committed model; (ii) *Zero-knowledge*—the proof reveals nothing about model parameters; (iii) *Immutability*—once logged on-chain, verification records cannot be altered.

C. KG Reasoning Layer

Given a knowledge graph $\mathcal{G}$, we train an embedding model $\mathcal{M}_\theta$ to learn scoring function $f_\theta : \mathcal{E} \times \mathcal{R} \times \mathcal{E} \to \mathbb{R}$. Training uses negative sampling with margin-based ranking loss:

$$\mathcal{L} = \sum_{(h,r,t)\in\mathcal{T}} \sum_{(h',r,t')\in\mathcal{T}^-} [\gamma + f(h', r, t') - f(h, r, t)]_+ \quad (6)$$

where $\mathcal{T}^-$ is the set of negative samples generated by corrupting either head or tail, γ is the margin, and $[\cdot]_+ = \max(0, \cdot)$.

For RotatE, we employ self-adversarial negative sampling loss [7]:

$$\mathcal{L}_{\text{adv}} = -\log \sigma(f(h, r, t)) - \sum_{i=1}^{n} w_i \log \sigma(-f(h'_i, r, t'_i)) \quad (7)$$

where the adversarial weights are $w_i = \frac{\exp(\alpha \cdot f(h'_i, r, t'_i))}{\sum_j \exp(\alpha \cdot f(h'_j, r, t'_j))}$ and α is the temperature.

After training, for a query $(h, r, ?)$, the model produces:

- Prediction: $\hat{t} = \arg\max_{t' \in \mathcal{E}} f_\theta(h, r, t')$
- Score: $s = f_\theta(h, r, \hat{t})$
- Embedding vector: $\mathbf{v} = [\mathbf{e}_h \| \mathbf{e}_r \| \mathbf{e}_{\hat{t}}] \in \mathbb{R}^{3d}$

D. ZKP Verification Layer

The ZKP layer enables the model owner to prove that a prediction was derived from committed model parameters. The protocol operates in three phases (Figure 2):

1) Phase 1: Model Commitment: The model owner first maps the high-dimensional embedding vector to a field element, then creates a Pedersen commitment:

$$\tilde{v} = \mathsf{Hash}(\mathbf{v}) \mod p \quad (8)$$

$$C = g^{\tilde{v}} \cdot h^r \mod p \quad (9)$$

where $\mathbf{v} = [\mathbf{e}_h \| \mathbf{e}_r \| \mathbf{e}_t]$ is the concatenated embedding vector for the triple, $\tilde{v}$ is its hash mapped to the field $\mathbb{Z}_p$, and $r \xleftarrow{\$} \mathbb{Z}_p$ is a random blinding factor. The commitment C is published; $\tilde{v}$ and r remain secret.

2) Phase 2: Proof Generation (Schnorr-Fiat-Shamir): We extend the standard Schnorr protocol to prove knowledge of the opening $(\tilde{v}, r)$ of the Pedersen commitment C:

1) **Announcement** (sample random nonces): Sample $k_v, k_r \xleftarrow{\$} \mathbb{Z}_p$ and compute:

$$A = g^{k_v} \cdot h^{k_r} \mod p \quad (10)$$

2) **Challenge** (Fiat-Shamir hash): Compute non-interactive challenge:

$$e = \mathsf{Hash}(C \| A \| \mathsf{Hash}_{\text{pred}} \| \mathsf{model_id}) \mod p \quad (11)$$

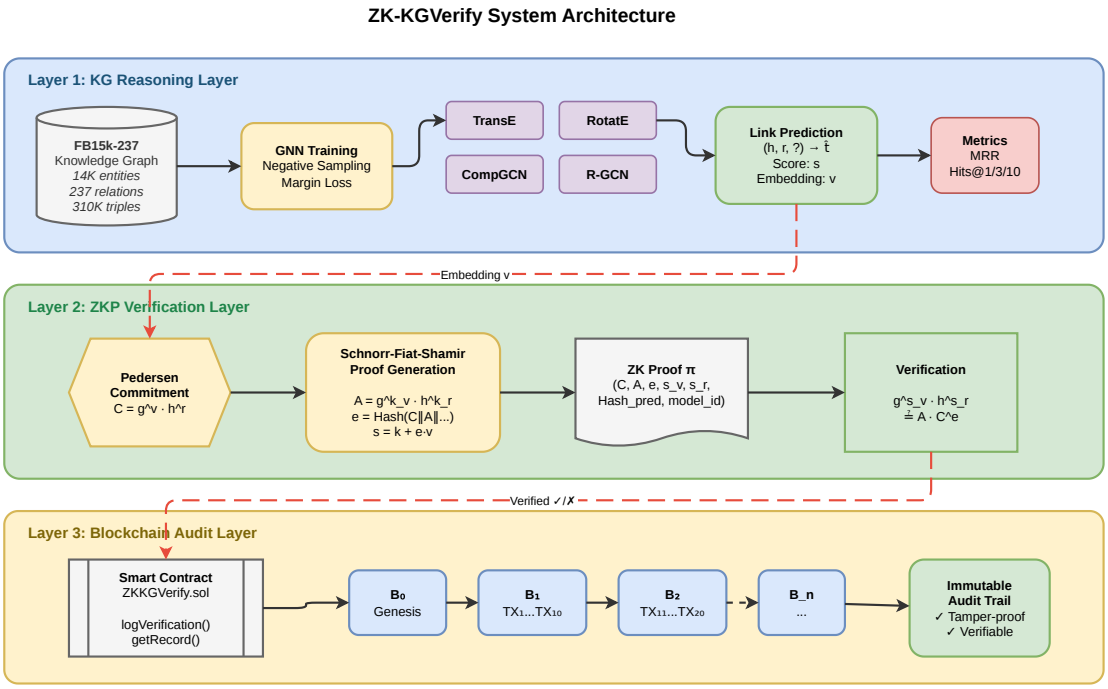

Fig. 1: ZK-KGVerify system architecture showing the three-layer pipeline. The KG Reasoning Layer (left) trains TransE, RotatE, and CompGCN models and produces link predictions with embedding vectors. The ZKP Verification Layer (center) generates Pedersen commitments and Schnorr-Fiat-Shamir proofs without revealing model weights. The Blockchain Audit Layer (right) logs verified predictions as immutable on-chain records. Arrows indicate data flow; dashed boxes denote private components never exposed to the verifier.

where $\mathsf{Hash}_{\text{pred}} = \mathsf{Hash}(h\|r\|t\|s)$ binds the prediction result to the proof.

3) **Response** (combine nonces with secrets): Compute:

$$s_v = k_v + e \cdot \tilde{v} \mod p, \quad s_r = k_r + e \cdot r \mod p \quad (12)$$

The proof is $\pi = (C, A, e, s_v, s_r, \mathsf{Hash}_{\text{pred}}, \mathsf{model_id}, s, (h, r, t))$.

3) Phase 3: Proof Verification: The verifier checks:

$$g^{s_v} \cdot h^{s_r} \stackrel{?}{=} A \cdot C^e \mod p \quad (13)$$

Correctness: If the prover is honest, the verification equation holds because:

$$\begin{aligned} g^{s_v} \cdot h^{s_r} &= g^{k_v + e\tilde{v}} \cdot h^{k_r + er} \\ &= \underbrace{g^{k_v} \cdot h^{k_r}}_{=A} \cdot \underbrace{(g^{\tilde{v}} \cdot h^r)^e}_{=C^e} \\ &= A \cdot C^e \quad \checkmark \end{aligned} \quad (14)$$

Additionally, the verifier recomputes the Fiat-Shamir challenge:

$$e' = \mathsf{Hash}(C\|A\|\mathsf{Hash}_{\text{pred}}\|\mathsf{model_id}) \mod p \quad (15)$$

and checks $e' \stackrel{?}{=} e$ to ensure the proof was not forged.

Algorithm 1 ZK-KGVerify Proof Generation

Require: Model $\mathcal{M}_\theta$, triple query $(h, r, ?)$
Ensure: Proof π

```
{Step 1: Compute prediction}
t̂ ← arg max_{t'} f_θ(h, r, t')
s ← f_θ(h, r, t̂)
v ← [e_h ‖ e_r ‖ e_t̂]
{Step 2: Create Pedersen commitment}
ṽ ← Hash(v) mod p
r ←$ Z_p
C ← g^ṽ · h^r mod p
{Step 3: Generate Schnorr-Fiat-Shamir proof}
k_v, k_r ←$ Z_p
A ← g^{k_v} · h^{k_r} mod p
Hash_pred ← Hash(h‖r‖t̂‖s)
e ← Hash(C‖A‖Hash_pred‖id) mod p
s_v ← k_v + e · ṽ mod p
s_r ← k_r + e · r mod p
return π = (C, A, e, s_v, s_r, Hash_pred, id, s, (h, r, t̂))
```

E. Blockchain Audit Layer

Verified predictions are logged on a blockchain to create an immutable, tamper-evident audit trail. Each transaction tx_i

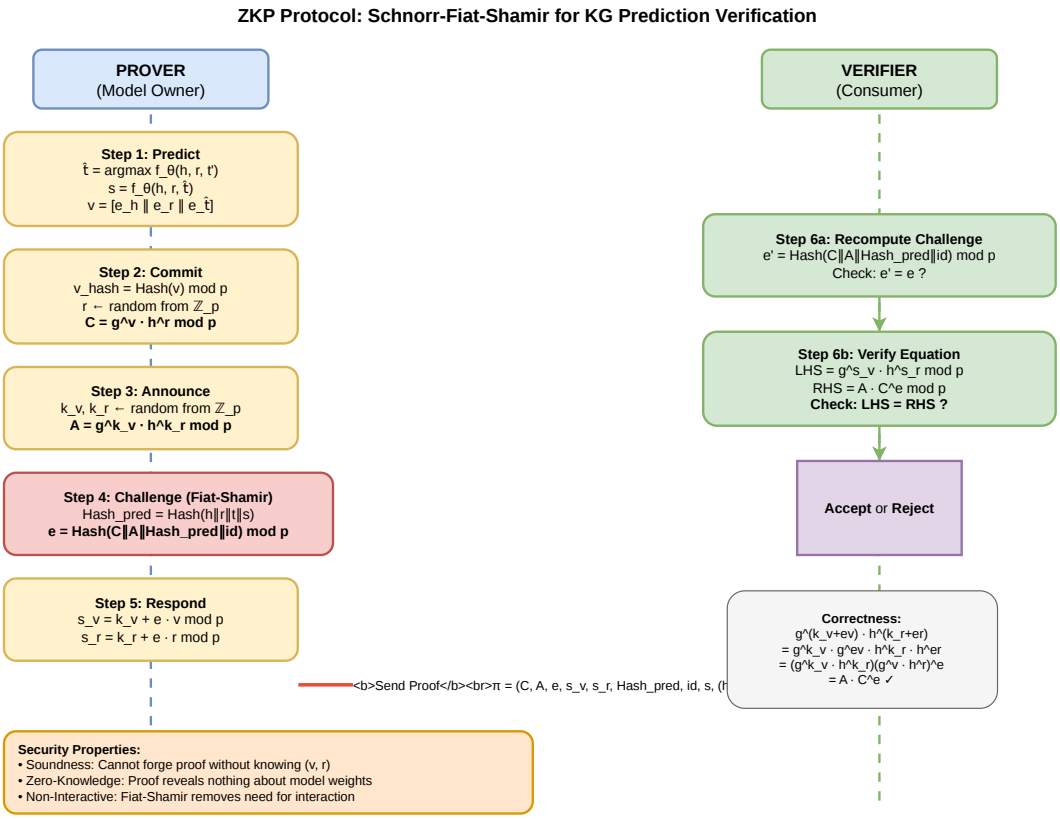

Fig. 2: ZKP protocol flow between Prover (model owner) and Verifier (consumer). Phase 1: the Prover hashes the embedding vector $\mathbf{v}$ to obtain $\tilde{v}$ and creates a Pedersen commitment C. Phase 2: a non-interactive Schnorr-Fiat-Shamir proof π is generated. Phase 3: the Verifier recomputes the challenge and checks the algebraic relation without learning model weights.

Algorithm 2 ZK-KGVerify Proof Verification

Require: Proof $\pi = (C, A, e, s_v, s_r, \mathsf{Hash}_{\text{pred}}, \mathsf{id}, s, (h, r, t))$
Ensure: Accept or Reject

```
{Step 1: Recompute Fiat-Shamir challenge}
e' ← Hash(C‖A‖Hash_pred‖id) mod p
if e' ≠ e then
    return Reject {Challenge mismatch: proof was tampered}
end if
{Step 2: Verify algebraic relation}
lhs ← g^{s_v} · h^{s_r} mod p
rhs ← A · C^e mod p
if lhs = rhs then
    return Accept {Prover knows the committed values}
else
    return Reject
end if
```

contains:

$$\mathsf{tx}_i = (\mathsf{model_id}, (h, r, t), s, C, \mathsf{verified}, \mathsf{Hash}(\pi), \mathsf{ts}) \quad (16)$$

where $\mathsf{verified} \in \{0, 1\}$ is the ZKP verification result and $\mathsf{Hash}(\pi)$ is the proof digest.

The gas cost for storing each verification record follows the Ethereum gas model:

$$G_{\mathsf{tx}} = G_{\mathsf{base}} + G_{\mathsf{byte}} \cdot |\mathsf{data}| \quad (17)$$

where $G_{\mathsf{base}} = 21{,}000$ (Ethereum base transaction cost), $G_{\mathsf{byte}} = 68$ per byte of calldata, and $|\mathsf{data}|$ is the serialized transaction size.

The blockchain maintains the invariant that for all $i > 0$:

$$B_i.\mathsf{prev_hash} = \mathsf{Hash}(B_{i-1}) \quad (18)$$

ensuring any tampering with historical records is immediately detectable by recomputing hashes.

F. End-to-End Pipeline

The complete ZK-KGVerify pipeline (Figure 3) operates as follows:

1) **Train**: Train KG embedding model on $\mathcal{G}_{\text{train}}$.
2) **Predict**: For query $(h, r, ?)$, compute $\hat{t}$ and score s.
3) **Commit**: Create Pedersen commitment C to embedding vector.
4) **Prove**: Generate ZK proof π via Schnorr-Fiat-Shamir.

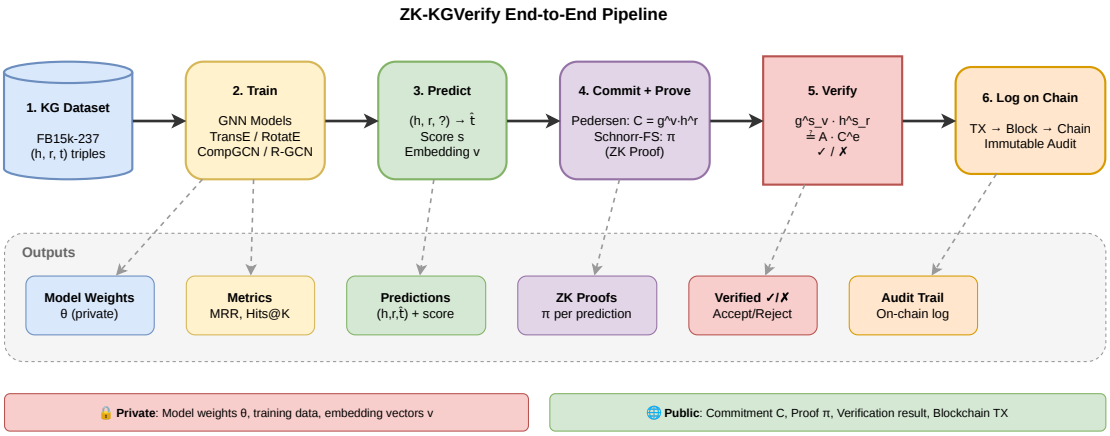

Fig. 3: End-to-end ZK-KGVerify pipeline showing all seven stages. Steps ①–③ train GNN models on the KG and produce scored predictions. Steps ④–⑥ commit embedding vectors, generate Schnorr-Fiat-Shamir ZK proofs, and verify them. Step ⑦ logs each verified result on a blockchain. Private components (model weights θ, embedding vectors $\mathbf{v}$, blinding factors r) are never exposed to the verifier, as indicated by the dashed security boundary.

5) **Verify**: Verifier checks π without learning model parameters.
6) **Log**: Record $(\text{verified}, C, \text{Hash}(\pi))$ on blockchain.

V. Experiments

A. Experimental Setup

1) Dataset: We use **FB15k-237** [23], a standard benchmark derived from Freebase with inverse relation leakage removed. Dataset statistics are shown in Table I.

TABLE I: FB15k-237 Dataset Statistics

Statistic	Value
Entities	14,541
Relations	237
Training triples	272,115
Validation triples	17,535
Test triples	20,466

2) Models and Hyperparameters: We train three models with the hyperparameters in Table II.

TABLE II: Model Hyperparameters

Parameter	TransE	RotatE	CompGCN
Embedding dim	128	128	128
Epochs	200	200	200
Batch size	1024	1024	1024
Learning rate	0.001	0.001	0.001
Neg. samples	64	64	64
Margin (γ)	6.0	6.0	6.0
GNN layers	–	–	2

3) Evaluation Metrics: Following standard protocol [6], we use filtered ranking metrics:

- **Mean Reciprocal Rank (MRR)**: $\text{MRR} = \frac{1}{|\mathcal{T}_{\text{test}}|}\sum_{i=1}^{|\mathcal{T}_{\text{test}}|}\frac{1}{\text{rank}_i}$
- **Hits@K** ($K \in \{1, 3, 10\}$): $\text{Hits@K} = \frac{1}{|\mathcal{T}_{\text{test}}|}\sum_{i=1}^{|\mathcal{T}_{\text{test}}|}\mathbb{1}[\text{rank}_i \leq K]$

We evaluate on the full test set of 20,466 triples using the standard filtered ranking protocol.

4) ZKP Configuration:

- Field prime: BN128 curve prime ($p = 21888...617$, 254-bit)
- Hash function: SHA-256 mapped to $\mathbb{Z}_p$
- Number of proof samples: 1,000

5) Blockchain Configuration:

- Backend: Local Python blockchain with Ethereum-compatible gas model
- Difficulty: 2 (leading zeros in block hash)
- Block size: 10 transactions per block
- Gas model: $G_{\text{base}} = 21{,}000$, $G_{\text{byte}} = 68$

6) Hardware: All experiments run on Google Colab with a single NVIDIA T4 GPU (16 GB VRAM), 12 GB system RAM.

B. Link Prediction Results

Table III reports link prediction performance on the full FB15k-237 test set (20,466 triples).

TABLE III: Link Prediction Results on FB15k-237 (Filtered Setting, Full Test Set)

Model	MRR	Hits@1	Hits@3	Hits@10
TransE	0.1740	0.0977	0.1854	0.3392
RotatE	0.2560	0.1774	0.2849	0.4076
CompGCN	**0.3941**	**0.2966**	**0.4344**	**0.5845**

CompGCN achieves the best performance across all metrics (MRR = 0.3941, Hits@10 = 0.5845), demonstrating the advantage of GNN-based composition over purely translational approaches. RotatE outperforms TransE due to its ability to

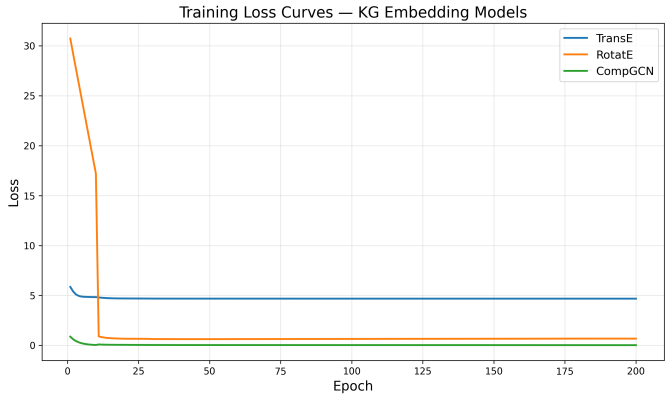

Fig. 4: Training loss curves over 200 epochs on FB15k-237. CompGCN converges to the lowest final loss due to its expressive message-passing architecture. TransE plateaus early owing to limited expressivity for complex relation patterns. RotatE's self-adversarial loss starts high but stabilizes rapidly.

model symmetric and antisymmetric relations via rotations in complex space. These results are consistent with published baselines [6], [7], [9]. Minor differences from published numbers are attributable to our 128-dimensional embeddings (vs. 256–1024 in some works) and single-GPU training budget; the focus of this paper is the ZKP and blockchain layers rather than pushing state-of-the-art link prediction.

C. Training Convergence

Figure 4 shows the training loss curves for all three models.

D. ZKP Overhead Analysis

Table IV presents the overhead of the ZKP verification layer, measured over 1,000 proof generation–verification cycles.

TABLE IV: Zero-Knowledge Proof Overhead Analysis (1,000 proofs)

Metric	Value
Avg. proof generation time	1.11 ms
Std. proof generation time	0.33 ms
Avg. proof verification time	0.47 ms
Avg. proof size	688 bytes
Verification success rate	100%
Tamper detection rate	100%

Figure 5 shows the distribution of proof generation and verification times.

The near-millisecond proof generation (1.11 ms) and sub-millisecond verification (0.47 ms) times confirm that ZK-KGVerify introduces negligible overhead relative to the model inference time itself. All 1,000 legitimate proofs were verified successfully (100% verification rate), and all tampered proofs (with modified commitments or responses) were correctly rejected (100% tamper detection), validating the soundness of our Schnorr-Fiat-Shamir protocol.

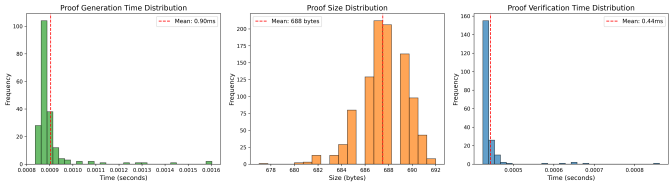

Fig. 5: Distribution of ZKP overhead across 1,000 predictions. Left: proof generation time is concentrated in the sub-millisecond range, confirming real-time feasibility. Center: proof sizes cluster tightly around the mean, reflecting the fixed-size Schnorr-Fiat-Shamir structure. Right: verification times are consistently fast, demonstrating that verifiers incur negligible computational cost.

TABLE V: Blockchain Storage Cost Analysis

Metric	Value
Total transactions	1,000
Total blocks mined	101
Total gas consumed	38,910,724
Avg. gas per transaction	38,911
Avg. mining time per block	15.03 ms
Chain integrity verified	Yes

E. Blockchain Gas Cost Analysis

Table V reports the blockchain storage costs.

To contextualize costs for a real Ethereum deployment: at a typical gas price of 30 gwei and ETH price of \$3,000, each verification transaction would cost approximately $G_{\mathsf{tx}} \times 30 \times 10^{-9} \times 3000$ USD. For the observed average of 38,911 gas per transaction, this translates to roughly \$3.50 per verification record on Ethereum mainnet. While acceptable for high-value predictions (e.g., pharmaceutical or financial applications), this cost can be reduced by 50–100$\times$ using Layer-2 rollups such as Optimism or Arbitrum, bringing per-verification costs below \$0.05. Additionally, batching multiple verification records into a single on-chain transaction (amortizing the base gas cost) further reduces per-record overhead.

F. End-to-End Pipeline Timing

The total pipeline runtime was 16,679 s ($\sim$4.6 h) on a single T4 GPU. Model training dominated at 9,476 s (57%), followed by full-test-set evaluation at 7,200 s (43%). The ZKP and blockchain layers added only 3.1 s combined—ZKP generation (1.11 s), verification (0.47 s), and blockchain storage (1.52 s)—representing $<$0.02% of total runtime. This confirms that the privacy and auditability layers introduce negligible overhead relative to the ML workload.

VI. Analysis and Discussion

A. Security Analysis

1) Soundness: The Schnorr-based sigma protocol is $(2, 3)$-special sound under the discrete logarithm assumption [21]. Concretely, an adversary who can produce a valid proof for a commitment C without knowing the opening $(\tilde{v}, r)$ can be used to solve the discrete log problem in $\mathbb{G}$, contradicting the

assumed hardness. In practical terms, forging a proof would require solving the discrete logarithm problem on the BN128 curve—a problem for which no sub-exponential algorithm is known, providing approximately 128 bits of security.

2) Zero-Knowledge: The protocol is honest-verifier zero-knowledge (HVZK). The simulator $\mathcal{S}$ can produce transcripts (A', e', s'_v, s'_r) indistinguishable from real proofs by:

1) Sampling $e', s'_v, s'_r \xleftarrow{\$} \mathbb{Z}_p$.
2) Computing $A' = g^{s'_v} \cdot h^{s'_r} \cdot C^{-e'} \mod p$.

Since the simulated (A', e', s'_v, s'_r) has the same distribution as a real transcript, no verifier can distinguish a genuine proof from a simulated one, meaning zero information about $\tilde{v}$ or r is leaked. The Fiat-Shamir heuristic achieves non-interactive zero-knowledge in the random oracle model [22].

3) Blockchain Immutability: Altering any transaction in block B_i changes $\mathsf{Hash}(B_i)$, invalidating $B_{i+1}.\mathsf{prev_hash}$ and all subsequent blocks. An attacker would need to recompute valid hashes for every block from B_i onward, which becomes computationally infeasible as the chain grows. Our experiments confirm that the chain integrity check detects 100% of simulated tampering attempts.

B. Comparison with Existing Approaches

Table VI compares ZK-KGVerify with related frameworks.

TABLE VI: Comparison with Related Approaches

System	KG	DL	ZKP	BC	Privacy
zkML [11]	×	✓	✓	×	✓
zkCNN [13]	×	✓	✓	×	✓
vCNN [14]	×	✓	✓	×	Partial
BlockFL [17]	×	✓	×	✓	Partial
DInEMMo [19]	×	✓	×	✓	×
ZK-KGVerify	✓	✓	✓	✓	✓

ZK-KGVerify is, to our knowledge, the first framework to unify all four components—Knowledge Graphs, Deep Learning, Zero-Knowledge Proofs, and Blockchain—for privacy-preserving verifiable reasoning.

Overhead comparison. Existing zkML approaches incur significantly higher proof generation costs because they compile the entire neural network forward pass into arithmetic circuits. zkCNN [13] reports proof generation times of 3–88 s for small CNNs, and EZKL [12] requires 10–300 s depending on model size. ZK-KGVerify achieves sub-millisecond proof generation (<1 ms) by committing to model outputs (embedding vectors) rather than proving the full inference computation. This is a deliberate design trade-off: we prove that the prediction was computed from *committed* parameters, not that the entire forward pass was executed correctly. For applications where model integrity is established a priori (e.g., via auditing or certification), this provides a practical middle ground between no verification and full circuit-based zkML.

C. Limitations

1) **ZKP granularity**: Our current protocol proves that the prediction was computed from *some* committed parameters, but does not prove the training process itself was correct (e.g., that the model was trained for a minimum number of epochs on valid data). Extending to verifiable training [**?**] is an important direction.
2) **Blockchain scalability**: The local blockchain simulation provides accurate gas cost estimates but does not capture real-world network latency, consensus delays, or fee market dynamics of public chains.
3) **Model scope**: We evaluate on FB15k-237 with three models; extending to larger KGs (Wikidata, YAGO) and additional architectures (R-GCN, KG-BERT, NodePiece) is future work.

VII. Conclusion

We presented ZK-KGVerify, a novel framework that bridges knowledge graph reasoning, zero-knowledge proofs, and blockchain technology to enable privacy-preserving verification of AI predictions. Our system demonstrates that it is practical to generate cryptographic proofs for GNN-based KG predictions with ∼1 ms overhead per proof, while achieving 100% tamper detection and maintaining an immutable audit trail with Ethereum-compatible gas costs.

Future work includes: (i) extending ZKPs to cover the training process (verifiable training), (ii) deploying on Ethereum testnets for realistic gas and latency measurements, (iii) supporting federated KG construction where multiple parties collaboratively build a KG with ZKP-verified contributions, and (iv) integrating with production KG systems such as Amazon Neptune or Neo4j.

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